6 A particle P moves in a straight line with position given by $s(t) = 80e^{-\frac{t}{10}} - 40t$ m, $t \geqslant 0$ seconds.

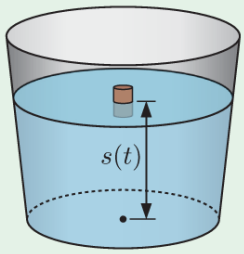
a Find the velocity and acceleration functions.

b Find the initial position, velocity, and acceleration of P.

c Sketch the graph of the velocity function.

d Find the exact time when the velocity is -44 m s^{-1}.

7 A cork bobs up and down in a bucket of water. The distance from the centre of the cork to the bottom of the bucket is given by $s(t) = 30 + \cos \pi t$ cm, $t \geqslant 0$ seconds.



a Find the cork's velocity at times $t = 0$, $\frac{1}{2}$, 1, $1\frac{1}{2}$, and 2 seconds.

b Find the time intervals when the cork is falling.

8 A skier is travelling down a hill. Her velocity after t seconds is given by $v(t) = \dfrac{\left(t^{1.1} + 3t\right)^{1.5}}{10}$ m s^{-1}.

a Find the velocity of the skier after 4 seconds.

b Write an expression for the acceleration of the skier after t seconds.

c Find the acceleration of the skier after 2 seconds.

d Use technology to find the total distance travelled by the skier in the first 10 seconds.

9 A feather is falling with velocity function $v(t) = -\frac{1}{24}t^3 - \frac{1}{12}t$ m s^{-1}. The feather is initially 2 metres above the ground.

a Find the displacement function $s(t)$.

b Find the time taken for the feather to reach the ground.

10 Tyson and Maurice are competing in a 100 m sprint. Tyson runs with speed $v_1(t) = 10(1 - e^{-1.25t})$ m s^{-1}, and Maurice runs with speed $v_2(t) = 10.5(1 - e^{-t})$ m s^{-1}, for $t \geqslant 0$ seconds.

a Who is travelling faster after 2 seconds?

b Find $\displaystyle\int_0^5 v_1(t)\,dt$, and interpret your answer.

c Find displacement functions $s_1(t)$ and $s_2(t)$ for each sprinter.

d Who is winning the race after 3 seconds?

e Show that Tyson completes the 100 m in approximately 10.8 seconds.

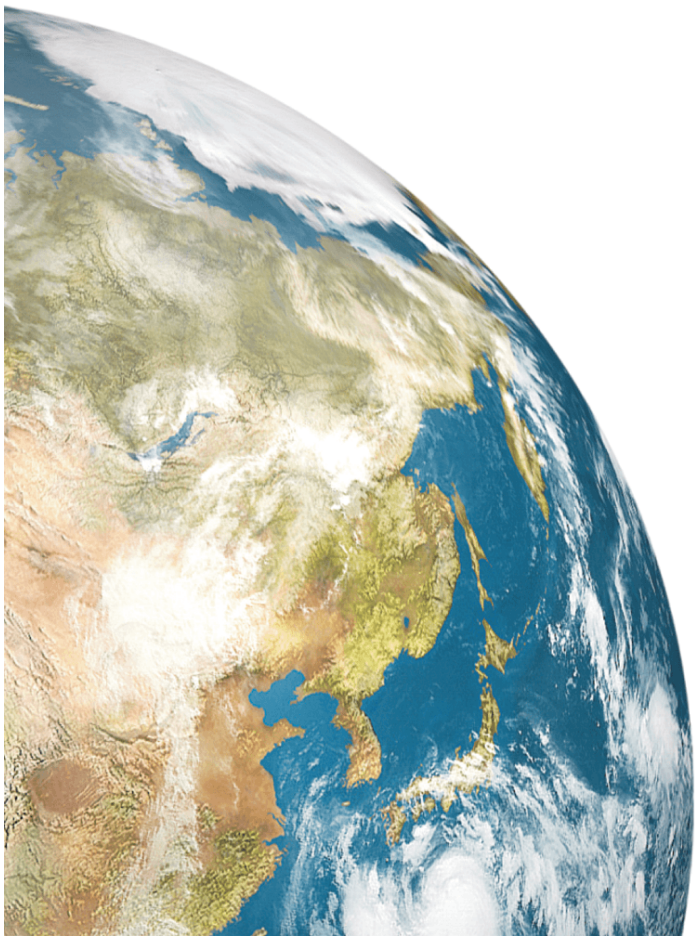
f Who will win the race?

Chapter 24

Maclaurin series

Contents:

A Maclaurin series
B Convergence
C Composite functions
D Addition and subtraction
E Differentiation and integration
F Multiplication
G Division



OPENING PROBLEM

Earlier in the course, we defined the *tangent* to a curve as the best straight line approximation to the curve at that point.

For a curve which is differentiable at $x = a$, the gradient of the tangent is $f'(a)$, and the tangent has equation $y = f(a) + f'(a)(x - a)$.

Consider the curve $f(x) = e^x$.

Things to think about:

a What is the tangent to the curve at $x = 0$?

b What is the best *quadratic* approximation to the curve at $x = 0$, having the same gradient and *curvature* (second derivative)?

c What is the best *cubic* approximation to the curve at $x = 0$?

d Could you continue this process to form an nth order polynomial approximation to the curve?

e Can we write an *exact* expression for e^x in the form $e^x = \sum_{k=0}^{\infty} c_n x^k$ where $c_n \in \mathbb{R}$?

f What uses could there be in having a polynomial approximation to a curve?

HISTORICAL NOTE

Prior to Archimedes, philosophers like **Zeno of Elea** and **Plato** had struggled with the mathematical idea that an infinite series could sum to a finite result. We have already seen several examples where Archimedes used an essentially infinite number of pieces to form a whole. Another is his approximation of π using the method of exhaustion to compute the areas of regular polygons with more and more sides.

In the 3rd century AD in China, **Liu Hui** independently developed similar methods. He made important commentary on the *Jiuzhang suanshu* or "Nine Chapters on the Mathematical Art" including early ideas about limits and infinitesimals, and calculations of volume formulae. Considering proof most important and modestly recognising when his own results were imperfect, he wrote "Let us leave the problem to whoever can tell the truth".



Liu Hui

In the 14th century, the Indian mathematician **Mādhava of Sangamagrāma** discovered infinite series expansions for a number of trigonometric functions, as well as infinite series for calculating π such as $\frac{\pi}{4} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2n+1}$.

The Scottish mathematician **James Gregory** (1638 - 1675) was probably unaware of Mādhava's work. He discovered infinite series representations for several trigonometric functions himself, and is regarded as the discoverer of the **Taylor series** representation for functions. He also published the first proof for the Fundamental Theorem of Calculus.

The Taylor series representation is named after **Brook Taylor** (1685 - 1731) who formalised the idea in 1715.

INVESTIGATION 1 TAYLOR SERIES

Consider a continuous function f which can be differentiated infinitely many times.

The Taylor series representation for f has the form

$$f(x) = \sum_{k=0}^{\infty} c_k(x-a)^k$$
$$= c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots.$$

This is a **power series** which generates an infinite polynomial representation for f.

What to do:

1 Explain why $c_0 = f(a)$.

2 Find $f'(x)$. Hence write c_1 in terms of $f'(a)$. What do you notice about the first two terms of the Taylor series expansion?

3 Find $f''(x)$. Hence write c_2 in terms of $f''(a)$.

4 **a** Prove that $f^{(n)}(x) = n!c_n + (n+1)!c_{n+1}(x-a) + \dfrac{(n+2)!}{2!}c_{n+2}(x-a)^2 + \ldots.$

$$= \sum_{k=0}^{\infty} \frac{(n+k)!}{k!}c_{n+k}(x-a)^k$$

b Hence write c_n in terms of $f^{(n)}(a)$.

c Write the Taylor series formula for f in terms of its function value and derivatives at $x = a$.

5 In the Taylor series we claim that a function can be represented everywhere using only information about the curve generated at $x = a$.

a Explain how information about the rest of the curve is established in the formula.

b Is it reasonable to expect that the sum of polynomial terms can exactly produce:

i a horizontal asymptote **ii** a vertical asymptote?

c Do you therefore expect that a Taylor series expansion will converge for *every* value of x in the domain of *every* function?

From the **Investigation** you should have established that

The **Taylor series expansion of $f(x)$ about $x = a$** is

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 \ldots. = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Functions defined in this way may look awkward. However, for values of x where they converge, power series can be added, differentiated, and integrated just like ordinary polynomials. This makes them extremely useful.

In this course we consider the special case of the Taylor series where $a = 0$. This case is called the **Maclaurin series**, named after **Colin Maclaurin** (1698 - 1746) who made extensive use of it in the 18th century.

A MACLAURIN SERIES

Consider a continuous function f which can be differentiated infinitely many times.
The **Maclaurin series expansion** of $f(x)$ is

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + ... = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)x^k}{k!}$$

For any values of x where the series converges, it is an exact representation of the function.

However, if we consider just the first n terms of the Maclaurin series expansion, we are left with an nth order polynomial approximation for the function.

The ***n*th degree Maclaurin polynomial** approximation to f is $M_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)x^k}{k!}$.

Example 1 **Self Tutor**

a Find the Maclaurin series representation for $f(x) = \frac{1}{2+x}$.

b Hence find $M_4(x)$, the 4th order Maclaurin polynomial approximation for $\frac{1}{2+x}$. Compare $M_4(1)$ with $f(1)$.

a
$$f(x) = \frac{1}{2+x} = (2+x)^{-1}$$
$$\therefore\ f'(x) = (-1)(2+x)^{-2}$$
$$\therefore\ f''(x) = (-1)(-2)(2+x)^{-3}$$
$$\vdots$$
$$f^{(k)}(x) = (-1)^k k!(2+x)^{-(k+1)}$$
$$\therefore\ f^{(k)}(0) = \frac{(-1)^k k!}{2^{k+1}} \text{ for all } k \in \mathbb{Z}^+.$$

Since $f(0) = \frac{1}{2}$, the Maclaurin series representation for $f(x)$ is

$$f(x) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{(-1)^k \cancel{k!}}{\cancel{k!}\,2^{k+1}} x^k$$
$$= \frac{1}{2} + \sum_{k=1}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k$$

b The 4th order Maclaurin approximation for $\frac{1}{2+x}$ is

$$M_4(x) = \sum_{k=0}^{4} \frac{(-1)^k}{2^{k+1}} x^k$$
$$= \frac{1}{2} + \frac{(-1)^1}{2^2}x + \frac{(-1)^2}{2^3}x^2 + \frac{(-1)^3}{2^4}x^3 + \frac{(-1)^4}{2^5}x^4$$
$$= \tfrac{1}{2} - \tfrac{1}{4}x + \tfrac{1}{8}x^2 - \tfrac{1}{16}x^3 + \tfrac{1}{32}x^4$$
$$\therefore\ M_4(1) = \tfrac{1}{2} - \tfrac{1}{4} + \tfrac{1}{8} - \tfrac{1}{16} + \tfrac{1}{32} = \tfrac{11}{32} \text{ compared with } f(1) = \tfrac{1}{3} = \tfrac{11}{33}.$$

EXERCISE 24A

1 Consider the function $f(x) = e^x$.

a Find the nth derivative $f^{(n)}(x)$ for $n \in \mathbb{Z}^+$.

Hence explain why $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + ...$

b The first two Maclaurin polynomial approximations for e^x are $M_1(x) = 1 + x$ and $M_2(x) = 1 + x + \frac{x^2}{2}$.

Write down the next *four* Maclaurin polynomial approximations.

c Graph $y = e^x$ and the first *four* polynomial approximations on the same set of axes. Discuss how the accuracy of the approximation is affected by the addition of the extra terms.

d Copy and complete this table of values:

n	1	2	3	4	5	6
$M_n(1)$						

Compare your results with $f(1) = e$.

2 Derive these Maclaurin series representations:

a $\frac{1}{1-x} = 1 + x + x^2 + x^3 + ... = \sum_{k=0}^{\infty} x^k$

b $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + ... = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k$

3 Explain why there is no Maclaurin series representation for $\ln x$.

4 Find an infinite series polynomial representation for $\frac{1}{x+1}$ using:

a the binomial theorem **b** Maclaurin series.

5 **a** Show that the Maclaurin series representation for $\cos x$ is $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ... = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$.

b Explain why there is no 3rd order Maclaurin polynomial in this case. What feature of $\cos x$ has caused this?

c Find $M_4(x)$, the fourth order Maclaurin polynomial approximation of $\cos x$.

Compare $M_4\left(\frac{\pi}{3}\right)$ with $\cos\frac{\pi}{3} = \frac{1}{2}$.

6 **a** Show that $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ... = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$.

b Explain why there is no 2nd order Maclaurin polynomial in this case. What feature of $\sin x$ has caused this?

c Find $M_5(x)$ and compare $M_5\left(\frac{\pi}{6}\right)$ with $\sin\frac{\pi}{6} = \frac{1}{2}$.

7 Derive these Maclaurin series representations:

a $\ln\left(\frac{1+x}{1-x}\right) = \sum_{k=1}^{\infty} \frac{2}{2k-1} x^{2k-1}$ **b** $\arctan x = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} x^{2k-1}$

INVESTIGATION 2 — $\int x^k\,dx$

We have previously seen that, for $k \neq -1$, $\int x^k\,dx = \frac{x^{k+1}}{k+1} + c$.

In this Investigation, we will consider the idea that a "better" formula might be $\int x^k\,dx = \frac{x^{k+1}-1}{k+1} + c$.

The expressions only differ by the constant $\frac{1}{k+1}$, and for $k \neq -1$ the derivative of both expressions is x^k. So, at first glance there may seem little meaningful difference between them.

To see why $\int x^k\,dx = \frac{x^{k+1}-1}{k+1} + c$ might be considered a "better" result, we will consider the limit of $\frac{x^{k+1}-1}{k+1}$ as $k \to -1$.

What to do:

1 We first establish the result $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$ discovered by **Leonhard Euler**.

a Show that in the expansion of $\left(1+\frac{x}{n}\right)^n$, the coefficient of x^i can be written as $\frac{1}{i!} \times \frac{n}{n} \times \frac{n-1}{n} \times \ldots \times \frac{n-i+1}{n}$.

b Hence find the limit of the coefficient of x^i as $n \to \infty$.

c Compare $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$ with the Maclaurin series expansion of e^x.

2 Suppose $y = \frac{x^{k+1}-1}{k+1}$ for $x > 0$.

a By letting $n = \frac{1}{k+1}$, show that $x = \left(1+\frac{y}{n}\right)^n$.

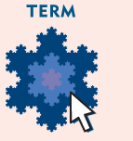
b Show that in the limit as $n \to \infty$, $x = e^y$.

c Hence show that $\lim_{k\to -1}\frac{x^{k+1}-1}{k+1} = \ln x$.

3 Can you explain why the formula $\int x^k\,dx = \frac{x^{k+1}-1}{k+1} + c$ might be considered "better" than $\int x^k\,dx = \frac{x^{k+1}}{k+1} + c$?

ACTIVITY — THE REMAINDER TERM

The **remainder term** is the difference between a function and its nth order Maclaurin polynomial approximation.



The remainder term is used to give us a maximum bound for the possible error in using the approximation. This is particularly useful if we do not have an accurate value to compare our approximation with.

B CONVERGENCE

DISCUSSION

Given your previous study of infinite geometric series, what condition(s) do you think might be necessary for the convergence of a Maclaurin series?

We have seen that the nth order Maclaurin polynomial approximation $M_n(x)$ can be used to approximate a function $f(x)$. We can say that $M_n(x)$ is the "nth order polynomial of best fit" for the curve.

Since $M_n(x)$ is generated using the values of f and its derivatives at $x = 0$, $M_n(x)$ will generally be most accurate near to $x = 0$.

As we get further from $x = 0$, we need to consider whether increasing the order of the polynomial will be sufficient to accurately model the shape of the function.

For example, it is not possible for any polynomial approximation to model a vertical asymptote, no matter what order polynomial is used.

The infinite Maclaurin series for $f(x)$ **converges** for any value of x such that

$$\lim_{n\to\infty} M_n(x) = f(x).$$

The set of values of x for which the infinite Maclaurin series converges is called the **interval of convergence**.

It should be clear that any interval of convergence for a Maclaurin series must include $x = 0$, since $M_n(0) = f(0)$ for all n.

From our previous study of infinite geometric series, we can also deduce that the infinite Maclaurin series will only converge for values of x such that the terms of the series are getting smaller.

For example, consider these series from **Exercise 24A**:

- $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots.$
  The terms decrease in size provided $|x| < 1$, and indeed the series is convergent for $|x| < 1$.

- $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \ldots.$
  It might seem like the fractions would help the terms to reduce in size. However, for any $|x| > 1$, the terms will certainly increase.
  For example, using l'Hôpital's rule, $\lim\limits_{n\to\infty} \dfrac{2^n}{n} = \lim\limits_{n\to\infty} \dfrac{2^n \ln 2}{1}$ which does not exist.
  For the case $x = -1$, $-1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \ldots.$ does not converge, which is reasonable since $\ln 0$ is undefined.
  For the case $x = 1$, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots.$ does converge.
  So, the Maclaurin series is convergent for $-1 < x \leqslant 1$.

- $e^x = 1 + x + \dfrac{1}{2}x^2 + \dfrac{1}{3!}x^3 + \dfrac{1}{4!}x^4 + \ldots.$
  In this case, the fractions *do* help the series to converge. For any value of x we pick, we can find a value of n for which $n!$ grows faster than x^n. Hence $\dfrac{x^n}{n!}$ will reduce in size. This series converges for all $x \in \mathbb{R}$.

EXERCISE 24B

1 Given the following Maclaurin series, discuss the values for which you think they converge:

a $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots.$

b $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots.$

c $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots.$

d $\ln\left(\frac{1+x}{1-x}\right) = 2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \frac{2}{7}x^7 + \ldots.$

2 Consider the binomial expansion for $(1+x)^p$, $p \in \mathbb{R}$.

a Show that the Maclaurin series for $(1+x)^p$, $p \in \mathbb{R}$, is $\sum_{k=0}^{\infty} \frac{p(p-1)\ldots(p-k+1)}{k!}\, x^k$.

Does this agree with the result previously stated as the **binomial theorem**?

b Explain why this series is only guaranteed to converge for $|x| < 1$.

3 **a** Show that the Maclaurin series for $\tan x$ is $\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \ldots.$

b Explain why this series is only convergent for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

4 **a** Show that the nth order Maclaurin polynomial for $\frac{1}{1-2x}$ is $M_n(x) = 1 + 2x + 4x^2 + 8x^3 + 16x^4 + \ldots. + 2^n x^n$.

b **i** Copy and complete this table:

n	1	2	3	4	5	6
$M_n(1)$						

ii Do you think this Maclaurin series converges when $x = 1$?

c **i** Copy and complete this table:

n	1	2	3	4	5	6
$M_n(0.1)$						

ii Do you think this Maclaurin series converges when $x = 0.1$?

d For what values of x do you think the Maclaurin series converges?

5 **a** Show that the Maclaurin series for $\sqrt{x+1}$ is $\sqrt{x+1} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \frac{7}{256}x^5 - \ldots.$

State the maximum domain on which this series *might* converge.

b Copy and complete the table:

n	1	2	3	4	5
$M_n(1)$					

Discuss whether you think the Maclaurin series is convergent for $x = 1$.

c Find the first six terms in the Maclaurin series for $\sqrt{x+1.8}$.

Using a similar table to **b** above, calculate $M_n(0.2)$ for the new series.

Compare the convergence of the two series when evaluating $\sqrt{2}$.

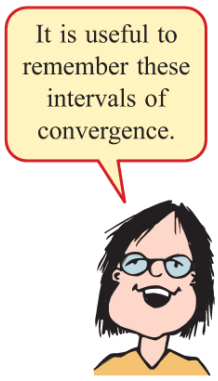
DISCUSSION

Suppose you want to use a Maclaurin polynomial $M_n(x)$ to approximate the value of a function at a particular value $x = a$.

Does it matter if the Maclaurin series is not *convergent* when $x = a$?

SUMMARY OF IMPORTANT MACLAURIN SERIES

Function	*Maclaurin series*	*Interval of convergence*
e^x	$1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \ldots.$	$x \in \mathbb{R}$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots.$	$x \in \mathbb{R}$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots.$	$x \in \mathbb{R}$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots.$	$\lvert x \rvert \leqslant 1$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots.$	$-1 < x \leqslant 1$
$(1+x)^p,\ \ p \in \mathbb{R}$	$\sum_{k=0}^{\infty} \frac{p(p-1)\ldots.(p-k+1)}{k!} x^k$	$\lvert x \rvert < 1$



C COMPOSITE FUNCTIONS

Having a standard list of common Maclaurin series is useful because we can use them to generate infinite series representations for other functions. We achieve this using **composite functions**.

Note that:

- the infinite series generated will only converge if the interval of convergence of the original Maclaurin series is obeyed
- the infinite series generated is only itself a Maclaurin series if it is a **polynomial**.

Example 2 Self Tutor

Use the Maclaurin series $\frac{1}{2+x} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k$ to find the Maclaurin series for $\frac{1}{2-x^2}$.

Write down the terms of the series up to x^6.

$$\frac{1}{2+x} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k$$

$$\therefore \frac{1}{2-x^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{k+1}} (-x^2)^k$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^{2k} x^{2k}}{2^{k+1}}$$

$$= \sum_{k=0}^{\infty} \frac{x^{2k}}{2^{k+1}}$$

$$= \frac{1}{2} + \frac{x^2}{4} + \frac{x^4}{8} + \frac{x^6}{16} + \ldots.$$

EXERCISE 24C

1 Use the Maclaurin series $\dfrac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ to find the Maclaurin series for the following.

In each case, write down the terms of the series up to x^4.

a $\dfrac{1}{1-x^2}$ **b** $\dfrac{1}{1+x}$ **c** $\dfrac{1}{1-2x}$ **d** $\dfrac{2}{1+x}$

2 Use the Maclaurin series $\cos x = \sum_{k=0}^{\infty} \dfrac{(-1)^k}{(2k)!} x^{2k}$ to find the Maclaurin series for the following.

In each case, write down the terms of the series up to x^6.

a $\cos 2x$ **b** $\cos(-x)$ **c** $\cos(x^2)$ **d** $x \cos x$

3 Use the Maclaurin series $\sin x = \sum_{k=0}^{\infty} \dfrac{(-1)^k}{(2k+1)!} x^{2k+1}$ to find the Maclaurin series for:

a $\sin(-x)$ **b** $\sin 3x$ **c** $\sin(x^2)$ **d** $x \sin x$

4 Use the Maclaurin series $e^x = \sum_{k=0}^{\infty} \dfrac{x^k}{k!}$ to find the Maclaurin series for the following.

In each case, write down the terms of the series up to x^4.

a e^{3x} **b** e^{x^2} **c** 2^x

5 **a** Use the Maclaurin series $\sin x = \sum_{k=0}^{\infty} \dfrac{(-1)^k}{(2k+1)!} x^{2k+1}$ to find an infinite series representation for $\sin\left(x + \frac{\pi}{2}\right)$.

b Use the binomial theorem to write out the terms of the series up to x^4. Explain your result.

Example 3 Self Tutor

a Use the Maclaurin series expansion for e^x to write down expansions for:

i e^{2x} **ii** e^{4x} **iii** e^{nx}, $n \in \mathbb{Z}^+$.

b Use the Maclaurin series expansion for $\cos x$ to find the first two non-zero terms of the Maclaurin series expansion of $\cos(e^x)$.

c Hence show that for x near 0, $\cos(e^x) \approx \cos(1) - \sin(1)x$.

d Find, by direct calculation, the Maclaurin polynomial of degree 2 of $\cos(e^x)$.

a $e^x = \sum_{k=0}^{\infty} \dfrac{x^k}{k!} = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \ldots$

i $e^{2x} = \sum_{k=0}^{\infty} \dfrac{(2x)^k}{k!} = 1 + 2x + \dfrac{4x^2}{2!} + \dfrac{8x^3}{3!} + \ldots$

ii $e^{4x} = \sum_{k=0}^{\infty} \dfrac{(4x)^k}{k!} = 1 + 4x + \dfrac{16x^2}{2!} + \dfrac{64x^3}{3!} + \ldots$

iii $e^{nx} = \sum_{k=0}^{\infty} \dfrac{(nx)^k}{k!} = 1 + nx + \dfrac{n^2x^2}{2!} + \dfrac{n^3x^3}{3!} + \ldots$

b $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \ldots$

$$\therefore \quad \cos(e^x) = 1 - \frac{e^{2x}}{2!} + \frac{e^{4x}}{4!} - \frac{e^{6x}}{6!} + \ldots.$$

$$= 1 - \frac{1}{2!}(1 + 2x + \frac{4x^2}{2!} + \ldots.) + \frac{1}{4!}(1 + 4x + \frac{16x^2}{2!} + \ldots.) + \ldots.$$

$$+ \frac{(-1)^k}{(2k)!}(1 + (2k)x + \frac{(2k)^2x^2}{2!} + \ldots.) + \ldots. \qquad \{\text{using a iii}\}$$

$$= 1 + \left(\sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!}\right) + \left(\sum_{k=1}^{\infty} \frac{(-1)^k(2k)}{(2k)!}\right)x + \ldots.$$

$$= 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k-1)!}\,x + \ldots.$$

c For x close to 0, higher order terms involving x^2, x^3, ... tend to 0.

$$\therefore \quad \cos(e^x) \approx 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k-1)!}\,x$$

Now $\cos(1) = 1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \ldots. = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!}$ {using known series for $\cos x$ and $\sin x$}

and $\sin(1) = 1 - \frac{1}{3!} + \frac{1}{5!} - \ldots. = -\sum_{k=1}^{\infty} \frac{(-1)^k}{(2k-1)!}$

$\therefore \quad \cos(e^x) \approx \cos(1) - \sin(1)x$

d Let $f(x) = \cos(e^x)$.

$\therefore \quad f'(x) = -\sin(e^x) \times e^x$

$\therefore \quad f''(x) = -e^x \sin(e^x) - \cos(e^x)e^{2x}$

$\therefore \quad f(0) = \cos(e^0) = \cos(1)$

$\therefore \quad f'(0) = -\sin(1) \times 1 = -\sin(1)$

$\therefore \quad f''(0) = -\sin(1) - \cos(1)$

$\therefore \quad M_2(x) = \cos(1) - \sin(1)x - (\sin(1) + \cos(1))\frac{x^2}{2!}$

6 **a** Use the Maclaurin series expansion for e^x to write down expansions for:

i e^{-x} **ii** e^{-3x} **iii** $e^{-(2n+1)x}$, $n \in \mathbb{Z}^+$.

b Use the Maclaurin series expansion for $\sin x$ to find the first two non-zero terms of the Maclaurin series expansion for $\sin(e^{-x})$.

c Hence find an approximation for $\sin(e^{-x})$ for x near 0.

d Find, by direct calculation, the Maclaurin polynomial of degree 2 of $\sin(e^{-x})$.

D ADDITION AND SUBTRACTION

Maclaurin series can be added and subtracted like normal polynomials. The resulting series will converge on the intersection of the intervals of convergence of the individual series.

EXERCISE 24D

1 Use $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots.$ and $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots.$ to write the terms of the Maclaurin series for the following up to x^7:

a $\sin x + \cos x$ **b** $\sin x + \sin 2x$ **c** $\sin(x^2) - \cos 3x$

2 **a** Use $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ to write the Maclaurin series for e^{-x}.

b The **hyperbolic sine function** is defined as $\sinh x = \frac{e^x - e^{-x}}{2}$.

Show that $\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$.

c The **hyperbolic cosine function** is defined as $\cosh x = \frac{e^x + e^{-x}}{2}$.

Find the Maclaurin series expansion for $\cosh x$.

3 Use the Maclaurin series expansion for e^x to find the series for $e^{i\theta}$ where $i = \sqrt{-1}$, $\theta \in \mathbb{R}$.

By separating the series into real and imaginary parts, prove **Euler's formula** $e^{i\theta} = \cos\theta + i\sin\theta$.

E DIFFERENTIATION AND INTEGRATION

One advantage of polynomial representations for functions is that they can be differentiated and integrated term by term.

This is particularly useful for integrating functions for which we otherwise cannot write an indefinite integral.

Example 4 **Self Tutor**

The Maclaurin series for $\cos x$ is $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$. Find the Maclaurin series for $\sin x$ by:

a differentiation **b** integration.

a $\frac{d}{dx}(\cos x) = -\sin x$

$\therefore \sin x = -\frac{d}{dx}\left(\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}\right)$

$= -\frac{d}{dx}\left(1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}\right)$

$= -\frac{d}{dx}(1) - \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k)!} \frac{d}{dx}(x^{2k})$

$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k)!} (2k) x^{2k-1}$

$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k-1)!} x^{2k-1}$

$= \sum_{n=0}^{\infty} \frac{(-1)^{n+2}}{(2(n+1)-1)!} x^{2(n+1)-1}$ {letting $n = k-1$}

$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$



b $\int \cos x\,dx = \sin x$

$\therefore \int_0^x \cos t\,dt = \big[\sin t\big]_0^x$

$= \sin x - \sin 0$

$= \sin x$

$\therefore \sin x = \int_0^x \left(\sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!}\right) dt$

$= \sum_{k=0}^{\infty} \left(\int_0^x \frac{(-1)^k t^{2k}}{(2k)!}\,dt\right)$

$= \left[\sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!}\right]_0^x$

$= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$ for all $x \in \mathbb{R}$

EXERCISE 24E

1 Show that the derivative of the Maclaurin series for e^x is itself.

2 The Maclaurin series of $\sin x$ is $\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$. Find the Maclaurin series for $\cos x$ by:

a differentiation **b** integration.

3 **a** Use the Maclaurin series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \ldots$ to find the Maclaurin series for $\frac{1}{1+x}$.

b *Hence* find the Maclaurin series for $\frac{1}{1+x^2}$.

c *Hence* show that the Maclaurin series for $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$

Example 5 **Self Tutor**

a Use the Maclaurin series $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$ to find the Maclaurin series for $\cos(x^2)$.

b Hence estimate $\int_0^1 \cos(x^2)\,dx$ correct to 4 decimal places.

a $\cos(x^2) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} (x^2)^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{4k}$

b $\int_0^1 \cos(x^2)\,dx = \int_0^1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{4k}\,dx$

$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \int_0^1 x^{4k}\,dx$

$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left[\frac{x^{4k+1}}{4k+1}\right]_0^1$

$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!\,(4k+1)}$

$= 1 - \frac{1}{10} + \frac{1}{216} - \frac{1}{9360} + \frac{1}{685\,440} - \ldots$ where $\frac{1}{685\,440} \approx 0.000\,001\,46$

Using the first 4 terms, $\int_0^1 \cos(x^2)\,dx \approx 0.9045$

4 **a** Use the Maclaurin series $\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$ to find the Maclaurin series for $\sin(x^2)$.

b Hence estimate $\int_0^1 \sin(x^2)\,dx$ correct to 4 decimal places.

5 **a** Use the Maclaurin series $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ to find the Maclaurin series for $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

b Hence estimate to 4 decimal places:

i $\int_{-1}^{1} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$ **ii** $\int_{-2}^{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$ **iii** $\int_{-3}^{3} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$.

6 **a** Use the Maclaurin series $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ to find the Maclaurin series for $\frac{1}{1+x^3}$. State the values of x for which you would expect the series to converge.

b Hence esimate $\int_0^{\frac{1}{3}} \frac{1}{1+x^3}\,dx$ to 4 decimal places.

Example 6

Self Tutor

a Use the Maclaurin series for $(1+x)^p$ to find the Maclaurin series for $(1-x^2)^{-\frac{1}{2}}$.

b Use the fact that $\arccos x - \arccos(0) = \int_0^x \frac{-1}{\sqrt{1-t^2}}\,dt$ to find a Maclaurin series for $\arccos x$. For which values of x is this valid?

c Hence find the first three non-zero terms of $\arccos(x^2)$, and use them to estimate $\int_0^{0.4} \arccos(x^2)\,dx$.

a $(1+x)^p = \sum_{k=0}^{\infty} \frac{p(p-1)\ldots(p-k+1)}{k!} x^k$ for all $|x| < 1$

$$\therefore (1-x^2)^{-\frac{1}{2}} = 1 + \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\ldots\left(\frac{1}{2}-k\right)}{k!}(-x^2)^k$$

$$= 1 + \sum_{k=1}^{\infty} \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\ldots\left(\frac{-2k+1}{2}\right)(-1)^k}{k!} x^{2k}$$

$$= 1 + \sum_{k=1}^{\infty} \frac{(-1)^{2k} \times 1 \times 3 \times 5 \times \ldots \times (2k-1)}{2^k\, k!} x^{2k}$$

$$= 1 + \sum_{k=1}^{\infty} \frac{1 \times 3 \times 5 \times \ldots \times (2k-1)}{2^k (k!)} x^{2k}$$

$$= 1 + \sum_{k=1}^{\infty} \frac{1 \times 3 \times 5 \times \ldots \times (2k-1)}{2^k k!} \times \frac{2 \times 4 \times 6 \times \ldots \times 2k}{2 \times 4 \times 6 \times \ldots \times 2k} \times x^{2k}$$

$$= 1 + \sum_{k=1}^{\infty} \frac{(2k)!}{2^k k!\, 2^k k!} x^{2k}$$

$$\therefore (1-x^2)^{-\frac{1}{2}} = 1 + \sum_{k=1}^{\infty} \frac{(2k)!}{4^k (k!)^2} x^{2k} \text{ for all } |x^2| < 1, \text{ which is for all } |x| < 1.$$

b $\arccos x - \arccos(0) = \displaystyle\int_0^x \frac{-1}{\sqrt{1-t^2}}\,dt$

$$\therefore \arccos x - \tfrac{\pi}{2} = -\int_0^x \left(1 + \sum_{k=1}^{\infty} \frac{(2k)!}{4^k(k!)^2}\,t^{2k}\right) dt$$

$$= -\left(x + \sum_{k=1}^{\infty} \frac{(2k)!}{4^k(k!)^2(2k+1)}\,x^{2k+1}\right)$$

$$\therefore \arccos x = \tfrac{\pi}{2} - x - \sum_{k=1}^{\infty} \frac{(2k)!}{4^k(k!)^2(2k+1)}\,x^{2k+1}$$

This is valid provided $|x| < 1$. These values of x all lie in the domain of $\arccos x$, so the series is valid for $-1 < x < 1$.

c $\arccos x \approx \frac{\pi}{2} - x - \frac{1}{6}x^3$ {using the first 3 terms}

$\therefore \arccos(x^2) \approx \frac{\pi}{2} - x^2 - \frac{1}{6}x^6$ for $|x| < 1$

$$\therefore \int_0^{0.4} \arccos\left(x^2\right) dx \approx \int_0^{0.4} \left(\tfrac{\pi}{2} - x^2 - \frac{x^6}{6}\right) dx \quad \text{since} \quad |0.4| < 1$$

$$= \left[\tfrac{\pi}{2}x - \frac{x^3}{3} - \frac{x^7}{7}\right]_0^{0.4}$$

$$\approx 0.607$$

7 **a** Use the Maclaurin series for $(1+x)^p$ to find the Maclaurin series for $(1+x^2)^{-1}$.

b Use the fact that $\arctan x - \arctan(0) = \displaystyle\int_0^x \frac{1}{1+t^2}\,dt$ to obtain the Maclaurin series for $\arctan x$. For which values of x is this valid?

c Hence find the first four non-zero terms of $\arctan(x^2)$, and use them to estimate $\displaystyle\int_0^1 \arctan(x^2)\,dx$.

8 **a** Use the Maclaurin series for $(1-x^2)^{-\frac{1}{2}}$ found in **Example 6** part **a**, and the fact that $\arcsin x - \arcsin(0) = \displaystyle\int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, to obtain the Maclaurin series for $\arcsin x$.

For which values of x is this valid?

b By considering $\arcsin\frac{1}{2}$, show that $\dfrac{\pi}{6} = \dfrac{1}{2} + \displaystyle\sum_{k=1}^{\infty} \frac{(2k)!}{2^{4k+1}(k!)^2(2k+1)}$.

F MULTIPLICATION

If we multiply two Maclaurin series together, we will be faced with collecting infinitely many "like" terms. This is generally not possible. However, by multiplying the first few terms of each Maclaurin series, we can generate the first few terms of the product.

Example 7 **Self Tutor**

The Maclaurin series expansion for $\cos 2x$ is $\cos 2x = 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + ...$

a Use the first few terms of $\sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - ....$ to find the terms of $\sin^2 x$ up to x^6.

b Compare your result with the expansion of $\frac{1}{2} - \frac{1}{2}\cos 2x$.

a
$$\begin{aligned}\sin^2 x &= \left(x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5 - ....\right)\left(x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5 - ....\right)\\ &= x^2 - \tfrac{1}{6}x^4 + \tfrac{1}{120}x^6 - ....\\ &\quad - \tfrac{1}{6}x^4 + \tfrac{1}{36}x^6 - ....\\ &\quad + \tfrac{1}{120}x^6 - ....\\ &= x^2 - \tfrac{1}{3}x^4 + \tfrac{2}{45}x^6 - ....\end{aligned}$$

b
$$\begin{aligned}\tfrac{1}{2} - \tfrac{1}{2}\cos 2x &= \tfrac{1}{2} - \tfrac{1}{2}\left(1 - 2x^2 + \tfrac{2}{3}x^4 - \tfrac{4}{45}x^6 + ...\right)\\ &= \tfrac{1}{2} - \tfrac{1}{2} + x^2 - \tfrac{1}{3}x^4 + \tfrac{2}{45}x^6 - ....\\ &= x^2 - \tfrac{1}{3}x^4 + \tfrac{2}{45}x^6 - ....\end{aligned}$$

which matches the first terms of the expansion of $\sin^2 x$.

EXERCISE 24F

1 By multiplication of the Maclaurin series for e^x and $\sin x$, find the first five non-zero terms in the Maclaurin series for $e^x \sin x$.

2 **a** Use the first few terms of $\sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - ....$ and $\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - ....$ to find the terms of $2\sin x \cos x$ up to x^5.

b Compare your result with the expansion of $\sin 2x$.

3 **a** Use $e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + ...$ and $\arccos x = \frac{\pi}{2} - x - \frac{1}{6}x^3 - ....$, $|x| < 1$ to find the first four non-zero terms in the Maclaurin series for $e^x \arccos x$.

b For which values of x will the Maclaurin series for $e^x \arccos x$ converge? Explain your answer.

4 **a** Use $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ to find the Maclaurin series for e^{-x^3}.

b Hence, or otherwise, find the terms in the Maclaurin series for $e^{-x^3}\cos x$ up to x^9.